

EASY TARGETS FOR COMPUTER VISION SYNDROME

98% professionals in urban India show symptoms of CVS

10% rise in complaints between 1999-2005

16 new patients on an average are treated every month

40 million Indians surf the Net and 180 million use cell phones every day

80% urban Indians use computers over 5 hours a day

NEW DEMANDS ON THE EYE

With computer screens fast supplanting print, eyes are forced to focus too close, too often, too long and too high...

- 1 DEMAND: Very fast and jumpy movements from one eye position to another—between document, screen and keyboard**

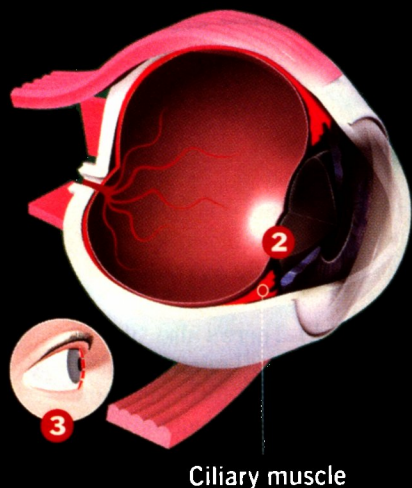
STRESS: Puts strain on the six muscles holding the globe and responsible for movements

- 2 DEMAND: Focusing too long, too close and too often**

STRESS: The ability of the eye to change focus is controlled by the ciliary muscle. Computer screen makes it work two-and-a-half times harder

- 3 DEMAND: Rate of blinking drops**

STRESS: Blinking spreads tear film over the eye, lubricating and protecting it. Reduced blinking causes the constituent layers in tear to break and evaporate faster. Result: dry, itchy, red and burning eyes



Ciliary muscle



over four hours a day. Three out of five of the 1.1 million IT workforce spend over 10 hours every day squinting at the screen (CyberMedia Dice-TNS report, 2007).

The medical fraternity is sitting up and taking note of the rising patient footfall. "Ten years back, CVS was unheard of. Today out of 12 patients a day, two to three would show computer-related eye strain," says optometrist, Ajeet Bhardwaj, who runs his clinic in Delhi. Dr C.P. Kamle has been tracking the disorder since 1999. As the director of Centre of Research in Environmental and Biological Sciences in Hyderabad, he had conducted the first scientific study of occupational health hazards on two lakh people in India. "Around 89 per cent showed symptoms of CVS," he

says. In 2005, in a spot survey on 134 ophthalmologists he found the figure to be a staggering 98 per cent. The latest issue of the *Indian Journal of Ophthalmology* surveys 300 ophthalmologists and reports that they average 16 CVS patients a month. Nearly 98 per cent patients show eye strain, 82 per cent headache, 79 per cent burning eye, 66 per cent watering, 61 per cent redness, 44 per cent shoulder pain and 36 per cent neck pain.

But why do computers cause such problems when reading books or papers for hours on end rarely does? "Normally eyes have little problem focusing on dense black, well-defined letters on a printed document, but the computer screen is different," explains Sachdev. "The blacks aren't as black

and whites aren't as white. Also the characters, called pixels, are brightest at the centre and diminish in intensity toward the edges. You can't focus well and your eyes get tired quicker."

Sorry, folks, holding down the control key on the keyboard and scrolling the mouse to make the font size ridiculously large doesn't really correct vision problems. "Working on a computer is really different than if you were reading a book or a newspaper," points out Dr Subrata Guhathakurta of West Bank Hospital in Kolkata. We are used to looking down to read, but focusing on a vertical computer system puts strain on the eye. "Position your monitor 16 to 30 inches away from your eyes," he advises. "The screen should be four to eight inches lower than eye-